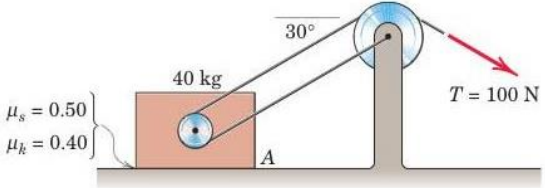
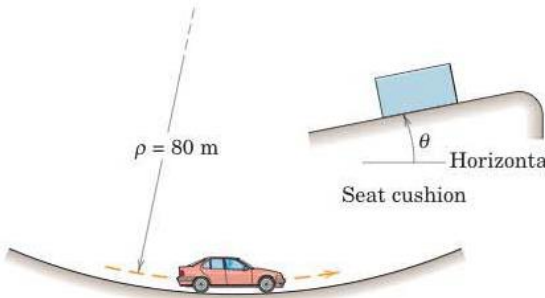
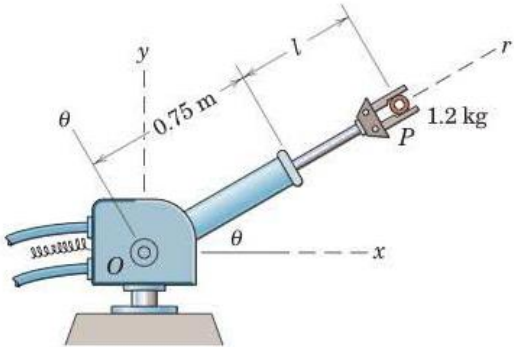
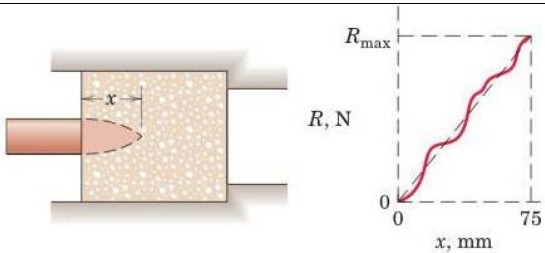
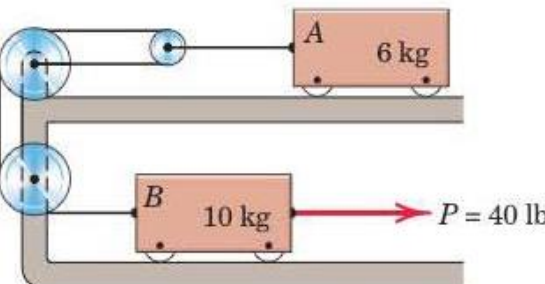


MEL1010 Engineering Mechanics

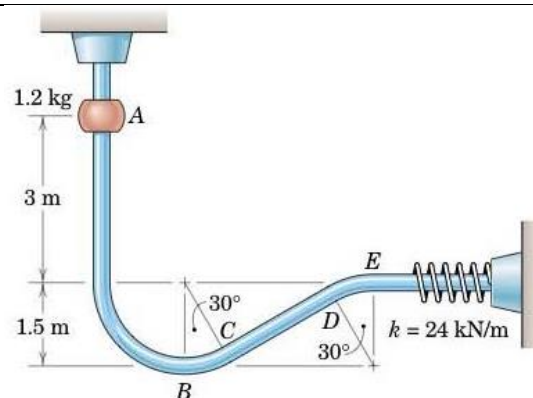
Tutorial 6 Work-Energy, Impulse Momentum and Impact

Answer all the following questions. Mention valid assumptions wherever applicable
(Note: 1-3 Newton's law, 4-6: Work Energy, 7-9: Impulse Momentum, 10-12: Impact)

1 Compute the acceleration of block <i>A</i> for the instant depicted. Neglect the masses of the pulleys. <i>Ans. $a = 1.406 \text{ m/s}^2$</i>	
2 The car has a speed of 70 km/h at the bottom of the dip when the driver applies the brakes, causing a deceleration of $0.5g$. What is the minimum seat cushion angle θ for which a package will not slide forward? The coefficient of static friction between the package and seat cushion is (a) 0.2 and (b) 0.4. <i>Ans. (a) $\theta = 7.34^\circ$, (b) $\theta = -3.16^\circ$</i>	
3 The robot arm is elevating and extending simultaneously. At a given instant, $\theta = 30^\circ$, $\dot{\theta} = 40 \text{ deg/s}$, $\ddot{\theta} = 120 \text{ deg/s}^2$, $l = 0.5 \text{ m}$, $\dot{l} = 0.4 \text{ m/s}$, and $\ddot{l} = -0.3 \text{ m/s}^2$. Compute the radial and transverse forces F_r and F_θ that the arm must exert on the gripped part <i>P</i>, which has a mass of 1.2 kg. Compare with the case of static equilibrium in the same position. <i>Ans. $F_r = 4.79 \text{ N}$, $(F_\theta)_{st} = 14.00 \text{ N}$ $(F_r)_{st} = 5.89 \text{ N}$, $(F_\theta)_{st} = 10.19 \text{ N}$</i>	
4 The resistance <i>R</i> to penetration <i>x</i> of a 0.25-kg projectile fired with a velocity of 600 m/s into a certain block of fibrous material is shown in the graph. Represent this resistance by the dashed line and compute the velocity <i>v</i> of the projectile for the instant when <i>x</i> = 25 mm if the projectile is brought to rest after a total penetration of 75 mm. <i>Ans. $v = 566 \text{ m/s}$</i>	
5 The force $P = 40 \text{ N}$ is applied to the system, which is initially at rest. Determine the speeds of <i>A</i> and <i>B</i> after <i>A</i> has moved 0.4 m. <i>Ans. $v_a = 1.180 \text{ m/s}$, $v_b = 2.36 \text{ m/s}$</i>	

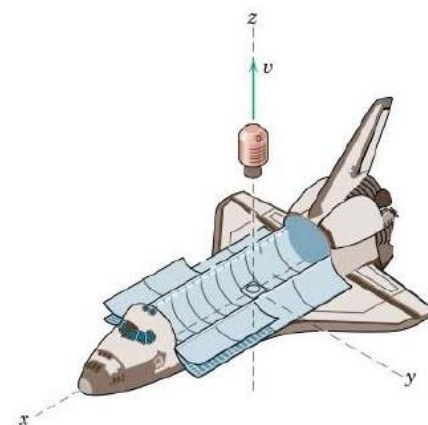
- 6 The 1.2-kg slider is released from rest in position A and slides without friction along the vertical-plane guide shown. Determine (a) the speed v_B of the slider as it passes position B and (b) the maximum deflection δ of the spring.

Ans. $v_b = 9.40$ m/s, $\delta = 54.2$ mm



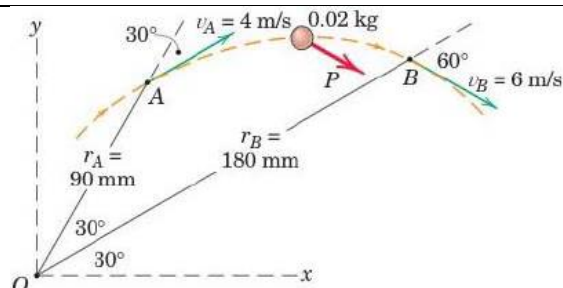
- 7 The space shuttle launches an 800-kg satellite by ejecting it from the cargo bay as shown. The ejection mechanism is activated and is in contact with the satellite for 4 s to give it a velocity of 0.3 m/s in the z-direction relative to the shuttle. The mass of the shuttle is 90 Mg. Determine the component of velocity v_f of the shuttle in the minus z-direction resulting from the ejection. Also find the time average F_{av} of the ejection force.

Ans. $v_f = 0.00264$ m/s, $F_{av} = 59.5$ N



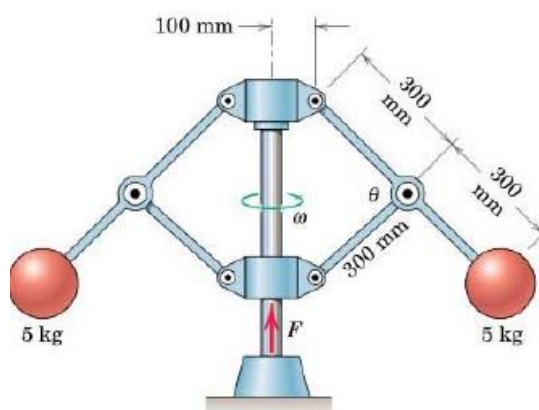
- 8 The 0.02-kg particle moves along the dashed trajectory shown and has indicated velocities at positions A and B. Calculate the time average of the moment about O of the resultant force P acting on the particle during the 0.5 second required for it to go from A to B.

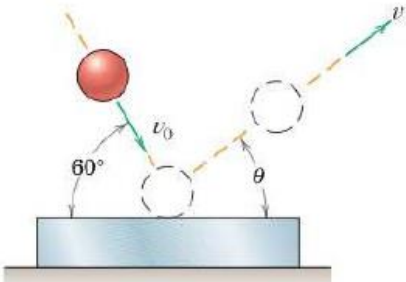
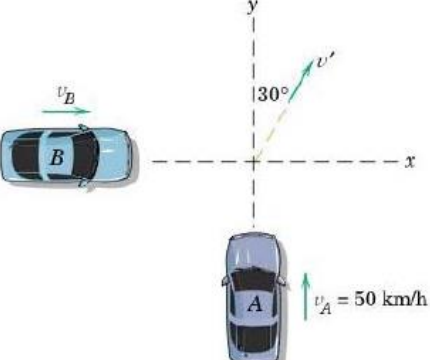
Ans. $Mo_{av} = 0.0302$ Nm



- 9 The assembly of two 5-kg spheres is rotating freely about the vertical axis at 40 rev/min with $\theta = 90^\circ$. If the force F which maintains the given position is increased to raise the base collar and reduce θ to 60° , determine the new angular velocity ω . Also determine the work U done by F in changing the configuration of the system. Assume that the mass of the arms and collars is negligible.

Ans. $\omega = 3$ rad/s, $U = 5.34$ J



1 0	The steel ball strikes the heavy steel plate with a velocity $v_0 = 24 \text{ m/s}$ at an angle of 60° with the horizontal. If the coefficient of restitution is $e = 0.8$, compute the velocity v and its direction θ with which the ball rebounds from the plate. <i>Ans. $\theta = 54.2^\circ, v = 20.5 \text{ m/s}$</i>	
1 1	The two cars collide at right angles in the intersection of two icy roads. Car A has a mass of 1200 kg and car B has a mass of 1600 kg. The cars become entangled and move off together with a common velocity v' in the direction initiated. If car A was travelling 50 km/h at the instant of impact, compute the corresponding velocity of car B just before impact. <i>Ans. $v_b = 21.7 \text{ km/h}$</i>	
1 2	Sphere A collides with sphere B as shown in the figure. If the coefficient of restitution is $e = 0.5$, determine the x- and y-components of the velocity of each sphere immediately after impact. Motion is confined to the x-y plane. <i>Ans. $v'_{Ax} = -1.672 \text{ m/s}, v'_{Ay} = 1.649 \text{ m/s}$ $v'_{Bx} = 6.99 \text{ m/s}, v'_{By} = -3.84 \text{ m/s}$</i>	